Project Initialization and Planning Phase

Date	16 October 2026
Team ID	
Project Title	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To automate and personalize travel planning using generative AI and route algorithms, delivering tailored, budget-aware day-by-day itineraries in seconds.
Scope	Covers user vibe capture, AI destination recommendation, daily schedule synthesis, 2-opt route transit optimization, and multi-format export (.ics/PDF).
Problem Statement	
Description	Travelers spend 10+ hours researching fragmented blogs and maps, often resulting in rigid, generic itineraries that cause severe daily transit backtracking.
Impact	Causes decision fatigue, wasted vacation time in traffic, budget overruns, and diminished traveler satisfaction.
Proposed Solution	
Approach	Combining Google Gemini 2.5 Flash for natural-language scheduling with a 2-opt TSP heuristic algorithm and Supabase PostgreSQL for cloud persistence.
Key Features	• AI vibe, pace, and budget-based daily itinerary synthesis.

	• 2-Opt TSP route optimizer to minimize travel transit time. • Real-time vibe sliders (+40% Adventure, Chill Mode). • 1-Click sync to Google/Apple Calendar (.ics) and printable PDF dossiers.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Dual-Core CPU / Render Cloud Container (0.5 CPU)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD (Local Dev) / 500 MB PostgreSQL (Supabase Cloud)
Software		
Frameworks	Python frameworks	FastAPI (Python backend), React 18 + Vite (Frontend)
Libraries	Additional libraries	google-genai, pydantic, uvicorn, haversine, icalendar, supabase
Development Environment	IDE	VS Code / Cursor IDE
Data		
Data	Source, size, format	Curated worldwide destinations dataset with coordinates and vibes, PostgreSQL / JSON format (schema.sql)